ASIA PACIFIC UNIVERSITY OF TECHNOLOGY AND INNOVATION
Level 2

CT149-3-2 - Software Testing
Assignment (100 Marks)

Learning Outcomes

1. Determine the aims, objectives, deliverables, and scope of a typical software testing approach.
2. Analyze the different types of software testing techniques for the respective purposes of these respective projects.
3. Select appropriate software testing tools to implement in the project.

Assignment Overview

- This coursework represents 50% of the entire course. It consists of;
  - Part-A Group Component (Marked as 80%)
  - Part-B Individual Component (marked as 20%)
- You are advised to attempt this coursework by forming a group of **not more than 6 members**. You are expected to collaborate and discuss your ideas among group members for all the questions.
- Submit both Individual and Group components in **ONE Final Report**, in-order of the question sequence stated below. All submission must be made to the Moodle System of your corresponding course link.
- Submission Date: **2 November 2026**

Case Study

ShopEasy Sdn Bhd operates a business-to-consumer e-commerce platform (web application and companion mobile app) that allows registered customers to browse products, manage a shopping cart, apply discount coupons, check out using multiple payment methods, and track their orders through to delivery. The back end is built as a set of microservices (Catalogue, Cart, Pricing/Promotions, Payment, Order, and Inventory) that communicate through REST APIs and integrate with a third-party payment gateway.

The development team follows Scrum, working in two-week sprints. Management has committed to releasing a new version of the platform — including a redesigned checkout flow and a new promotions engine — three weeks before the company's flagship “Black Friday” sale.

The QA team consists of four testers who are responsible for functional, regression, and acceptance testing before the release is signed off.

Given the compressed schedule and the business-critical nature of the sale, the Test Lead has been asked to plan the testing effort carefully and to consider which activities can be automated.

The key functional modules and business rules that testers must consider are summarized below.

1**. Registration and Login**

- Username must be 6–20 alphanumeric characters.
- Password must be 8–16 characters long and must contain at least one uppercase letter, one lowercase letter, one digit, and one special character (e.g., !@#$%).
- An account is locked for 30 minutes after 3 consecutive failed login attempts within a 15-minute window.

2. Product Search and Catalogue

- Customers can search by keyword, filter by category, and filter by price range.
- Search results are displayed 20 products per page, sorted by relevance, price, or customer rating.
- A product is automatically hidden from search results once its stock quantity reaches 0.

3. Shopping Cart

- A cart may contain a maximum of 10 distinct product lines.
- The maximum quantity per product line is 99 units.
- An inactive cart (no changes for 30 minutes) is automatically cleared.

4. Promotions and Coupon Engine

The Pricing/Promotions service applies to a discount at checkout based on the following business rules:

- Coupon “SAVE10” gives 10% off when the order subtotal is at least RM50.
- Coupon “SAVE20” gives 20% off when the order subtotal is at least RM150.
- Coupon “FREESHIP” waives the delivery fee when the order subtotal is at least RM80.
- Coupons SAVE10, SAVE20, and FREESHIP cannot be combined with one another — only one may be applied per order.
- New customers (account age ≤ 30 days) automatically receive an additional 5% new-customer discount, which CAN be stacked on top of SAVE10 only, provided the order subtotal exceeds RM50. It cannot be stacked with SAVE20 or FREESHIP.
- An invalid, expired, or already-used coupon code is rejected with an on-screen error message, and no discount is applied.

5. Checkout and Payment

- Accepted payment methods are: Credit/Debit Card, e-Wallet, and Cash on Delivery (COD).
- COD is only offered for orders with a total value below RM500; for orders of RM500 or more, only Card or e-Wallet may be used.
- Card payment requires a 16-digit card number, an expiry date that must be later than the current month/year, and a 3-digit CVV.
- On payment failure, the order is placed in a “Payment Failed” state and the customer may retry payment up to 3 times before the order is automatically cancelled.

6. Order Status Workflow

- Once payment is confirmed, an order moves through the following states, tracked by the Order service:

- Order Placed → Payment Confirmed → Processing → Shipped → Out for Delivery → Delivered.
- If payment fails, the order moves from “Order Placed” to “Payment Failed”, and then either back to “Order Placed” (on retry) or to “Cancelled” (after 3 failed retries).
- A customer may cancel an order at any point up to, but not including, the “Shipped” state. Orders that are “Shipped” or later cannot be cancelled.
- A customer may request a return only from the “Delivered” state, and only within 7 days of delivery; this moves the order to a “Return Requested” state.

7. Inventory Management (Admin)

- The system raises a low-stock alert to administrators when a product's stock quantity falls below 10 units.
- Administrators can adjust stock levels manually and configure supplier reorder rules.

Assignment Questions

PART-A – GROUP COMPONENT (80%)

1) Introduction

Include definition of testing and summary of available testing techniques.

[10 marks]

2) Functional and Non-Functional Requirements

Identify functional and non-functional requirements for Shop Easy. While students are encouraged to specify custom requirements tailored to the project scope, the baseline threshold requires at least 30 functional requirements and 10 non-functional requirements.

[10 marks]

3) Test techniques - Equivalent Partitioning

With the information you have collected from the case study, identify and create the Equivalent Partitioning (EP) diagram for password management function. Create test cases based on the identified EP diagram. **[10 marks]**

4) Test techniques – Boundary Value Analysis

Based on password management, design and create the value 2 Boundary Value Analysis(2 BVA) and 3 Boundary Value Analysis (3 BVA) for the given discount criteria.
Identify the coverage items and derive test cases for the identified diagram. **[15 marks]**

5) Test Cases and Traceability Matrix

Produce test cases for the software product to be tested mapped to the case study requirements. Use template of RTM provided in lecture. **[20 Marks]**

6) Test Tools

List and explain the functions and purpose of the TWO (2) test tools based on the test process activities that you would consider for Shop Easy. Provide examples of existing test tools in a real-time environment. **[15 Marks]**

PART-B – INDIVIDUAL COMPONENT (20%)

7) Propose one test design technique suitable for designing test cases during the system testing phase. Detail the step-by-step methodology for executing this technique within the context of the ShopEasy project.
Guidelines & Constraints:
Collaborative Discussion: Group members must propose, evaluate, and discuss various test design techniques prior to finalizing their selection.
Excluded Techniques: Equivalence Partitioning (EP), 2-value Boundary Value Analysis (2-BVA), and 3-value Boundary Value Analysis (3-BVA) are strictly prohibited for this task.

[20 marks]

Assignment Deliverables

- Final Documentation must be word-processed; submission is done online through Moodle.

- Include a 'Workload Matrix' (to be given by the lecturer), indicating the contribution of each individual for each required component (shown in percentage), and should be signed off by each team member, attached to the APPENDIX part of the final document.
- Citation of facts is mandatory. Obtain your facts from credible sources in references/bibliography. Avoid 'dumping of data'. Instead, the facts that you discuss should be made relevant to your case/project. Kindly use the **7th Edition of APA referencing style**.
- Late submissions will not be assessed unless extenuating circumstances are upheld.
- Font setup: use 1.5 spacing, justify paragraph alignment, 12-point font size, Times New Roman.
- The due date of this assignment is **2 November 2026**.

Performance Criteria

Grade	Assessment Guidelines
General	Pass answers are expected to be legible, tidy, well organised and written in clear, understandable English. Students who grossly exceed the word limit will be penalized.
0-49%	Superficial analysis, concepts and language of the subject are absent or scant. Irrelevant regurgitation of text book. Ideas are poorly expressed. Many key issues are ignored. Concepts and language of the subject are used but are often confused in application and or explanation. Some key issues are ignored.
50-64%	Some understanding of the relevant models and concepts. Some elements of an appropriate structure are present. Restricted analysis of some issues.
65-74%	Evidence of reading and research. Understanding of the application of appropriate models and concepts is demonstrated. Key issues are identified and analysed, although this may be restricted at times. Some sources are acknowledged.
75-79%	Evidence of wider reading. The assignment effectively interprets the information and exhibits the integration of ideas across the subject area. The assignment has credible recommendations. A systematic approach to development and evaluation is used. Most sources are acknowledged and referenced using APA style.
80% and above	Arguments are clear and convincing. Confident integration of theory and practices is demonstrated. Consistent referencing to sources using the APA style.